Program: **Diploma in Automation and Robotics**	
Course Code: **6331B**	Course Title: **Advanced Microprocessors and Microcontrollers**
Semester: **6**	Credits: **4**
Course Category: **Program core**	
Periods per week: **4 (L:4,T:0,P:0)**	Periods per semester: **60**

Course Objectives:

- To learn the architecture, programming and interfacing of microcontrollers with peripherals.
- To design microcontroller based embedded systems

Course Prerequisites:

- Microcontroller based Robotic Application Design

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To understand the Architecture of Intel 8086.	14	Understanding
CO2	To understand the Architecture of 80386.	12	Applying
CO3	To understand PIC18 architecture	16	Applying
CO4	To understand ARM architecture	16	Applying
	Series Test	**2**	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped 2-Moderately mapped 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To understand the Architecture of Intel 8086.**		
M1.01	To list the main features of Intel 8086	2	Understanding.
M1.02	To explain the internal architecture of Intel 8086	3	Understanding.
M1.03	To explain memory segmentation in 8086	3	Understanding.
M1.04	To describe the register set of 8086.	3	Understanding.
M1.05	To illustrate the Flag register in 8086.	3	Understanding.
Contents: Features of Intel 8086. The internal architecture of Intel 8086.memory segmentation in 8086. physical address generation in 8086. register set of 8086.Flag register in 8086. pin functions of 8086.Minimum mode and maximum mode configurations of 8086.			
CO2	**To understand the Architecture of 80386**		
M2.01	To list the key features of Intel 80386.	6	Understanding.
M2.02	To explain the internal architecture of 80386.	6	Understanding.
	Series Test – I	1	
Contents: Key features of Intel 80386 - internal architecture of 80386 - operating modes - paging mechanism - address translation in PVAM (non paged and paged modes) - features of Pentium processor - internal architecture of Pentium processor - list of operating modes			
CO3	**To understand PIC18 architecture**		
M3.01	Understand the PIC18 architecture	4	Understanding
M3.02	Understand Assembly language programming, I/O Port programming	4	Understanding
M3.03	LCD and Keyboard Interfacing,	3	Applying
M3.04	ADC and DAC interfacing.	3	Applying
M3.05	Sensor interfacing.	2	Applying
Contents: PIC18 architecture , Assembly language programming, I/O Port programming, addressing modes, Instruction set, programming in C,LCD and Keyboard Interfacing,			

ADC, DAC, and sensor interfacing.			
CO4	**To understand ARM architecture**		
M4.01	Understand the ARM architecture	4	Understanding
M4.02	Describe ARM organization and Implementation,	4	Understanding
M4.03	Describe Memory Hierarchy,	4	Applying
M4.04	ARM Instruction Set	4	Applying
	Series Test – II	1	
Contents: ARM architecture, ARM organization and Implementation, Memory Hierarchy, ARM Instruction Set and Thumb Instruction set, Assembly Language Programming.			

Text / Reference

T/R	Book Title/Author
T1	Microprocessor 8086 Programming & Interfacing - A NagoorKani - RBA Publications.
T2	Microprocessors and Interfacing - Douglas V Hall – TMH.
T3	The x86 Microprocessors - Second Edition - Lyla B Das – Pearson.
T4	Muhammad Ali Mazidi, Rolin D. Mckinlay, Danny Causey, “PIC Microcontroller and Embedded systems”, Pearson Education,2007
R1	S. Furber, “ARM System-on-Chip Architecture”, Pearson Education,2nd 2013.
R2	Andrew N. Sloss, Dominic Symes, Chris Wright, “ARM System Developer’s guide”, Morgan Kaufman,2004.
R3	David Seal, “ARM Architecture Reference Manual”, second edition Addison Wesley,2nd edition 2001
R4	Joseph Yiu, “The Definitive Guide to the ARM Cortex- M3”, Newness.2nd edition 2010
R5	PIC18FXX8 data sheets.
R6	Lyla B. Das, “Embedded systems, an integrated approach”,Pearson Education 2013